AN EXPLICIT FOUR-POINT RATIONAL DESIGN COUNTEREXAMPLE TO AIM QUESTION 47 IN ALGEBRAIC STATISTICS

ARGUS OPEN RESEARCH PROJECT

Abstract. Question 47 of the AIM problem list in computational algebraic statistics asks whether reducing the minimal generators $m_1, \dots, m_r$ of a monomial ideal $M \subset \mathbb{Q}[x_1, \dots, x_n]$ modulo a finite design $D = \{v_1, \dots, v_N\} \subset \mathbb{Q}^n$ against its standard monomial basis $O = \mathrm{Std}(M)$ yields polynomials $f_1, \dots, f_r \in I(D)$ whose common zero locus $V(f_1, \dots, f_r)$ equals D. We give an explicit four-point rational counterexample in $\mathbb{Q}[x, y]$ where $V(f_1, f_2)$ contains the entire line $x = y$.

1. The Four-Point Counterexample

In $\mathbb{Q}[x, y]$, define the four-point rational design and monomial ideal

$$D = \{(0,0),\ (1,1),\ (2,2),\ (0,1)\} \subset \mathbb{Q}^2, \qquad M = \langle x^2, y^2 \rangle, \qquad O = \mathrm{Std}(M) = \{1, x, y, xy\}.$$

Theorem 1.1. *For (D, M, O) above:*

(1) *Evaluating $(1, x, y, xy)$ on the four points of D yields the 4×4 matrix with determinant $-2 \neq 0$, so O is a basis of $\mathbb{Q}[x, y]/I(D)$.*

(2) *The unique interpolants of the minimal generators $m_1 = x^2$ and $m_2 = y^2$ in $\mathrm{span}_{\mathbb{Q}} O$ on D are $x^2 \equiv xy$ and $y^2 \equiv xy - x + y \pmod{I(D)}$, producing*

$$f_1 = x^2 - xy = x(x - y), \qquad f_2 = y^2 - xy + x - y = (y - 1)(y - x).$$

(3) *The common zero locus is $V(f_1, f_2) = \{(t, t) : t \in \mathbb{Q}\} \cup \{(0, 1)\} \supsetneq D$; in particular $(3, 3) \in V(f_1, f_2) \setminus D$.*

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